

Tutorial 2: Project Planning, Control and Process Models

Question 1

Case Study: Beamscope Canada, Inc.

Beamscope Canada, Inc. is an electronics distributor located in Scarborough, Ontario. In their manual order processing system, when a customer called to place an order, the clerk would scribble it down, run down the hall to the credit check guys and flip through files and folders to see if their credit was okay. The clerk would then run to see whether Beamscope actually had the item in stock and then go back to the telephone to confirm the order. There were two PCs at the company and they were doing \$93 million worth of business. The phone orders were manually routed to the warehouse, where they would sit for at least a week. Plus, it was hard to know what was actually in the warehouse and where it was. It took Beamscope five to seven days to ship an order and very often, there were mistakes made with the orders. In addition, products were collecting dust at Beamscope's physical warehouses. It also took a long time to compile reports for senior management and by the time the reports were printed, the information was already outdated. Senior management at Beamscope envisioned a new system that not only takes on order entry but also features a radio frequency bar-code system for inventory control, electronic data interchange and a data warehouse for decision support. In the new system, customers can place orders themselves, check the status of previous orders and download product literature. If an item is temporarily out of stock, the system will suggest a substitute or put the customer on a list for back orders. The system will also allow users to browse 120,000 square feet of shelves holding 10,500 different products in the company's 2 warehouses. In addition, the new system is directly tied into United Parcel Service, Inc.'s electronic shipping system.

a) Well-defined objectives should be specific, measureable, achievable, relevant, and time- constrained (SMART). Consider an objective: **design an user friendly system** for Beamscope's project to develop a new system that automates their order processing, inventory and warehousing.

Does this project objective fulfill the SMART criteria? Explain your answer.

b) Identify and describe TWO (2) stakeholders of the Beamscope's project.

1a)

SMART

Project Planning

Well-defined objectives should be:

- **Specific** – The project objective should be clear, well-defined, and unambiguous.
To improve overall cust. satisfaction rates to 8/10 by 30-Jun is a clear, well-defined and unambiguous objective
- **Measurable** – there should be measures of effectiveness which tell us how successful is the project.
- **Achievable** – it must be within the power of the individual or group to achieve the objective
** Cust satisfaction target = 8/10: achievable?
* If target = 10/10: achievable?*
- **Relevant** – the objective must be relevant to the true purpose of the project.
*- Project is "something" that you will do/carried out
- Objective is the "desired outcome" that you want to achieve*
- **Time constrained** – there should be a defined point in time by which the objective should have been achieved.
30-Jun is the deadline to achieve the desired outcome

SMART criteria	Yes/No	Evaluation
Specific	No	Not specific enough. It doesn't define what "user-friendly" means — e.g., easy navigation, intuitive interface or mobile compatibility
Measurable	No	Not measurable. There's no way to measure how "user-friendly" the system is. A measurable goal would include user satisfaction scores or task completion times.
Achievable	Yes	Achieve, assuming the team has the right skills and resources
Relevant	Yes	Relevant to the project's purpose— improving the efficiency of order processing and inventory

		control
Time-Constraint	No	The time-frame is not mentioned, there is no deadline for the development of the project.

Suggested Objective:

To design and implement a user-friendly automated order processing and inventory system within 6 months, which reduces order processing time by 70% and achieves at least 85% user satisfaction in post-deployment surveys.

1b)Stakeholders of Beamscope's project are:

The senior management envisioned the new system as a project that features storing order entry, and radio frequency bar-code system for inventory control, electronic data interchange and a data warehouse for decision support.

The second stakeholder group would be customers who request for the system's ability to place orders, check previous order statuses, and download product literature.

Other acceptable answers :

Clerk (line 2 in case study);

United Parcel Service Company (2nd last line in case study)

Team 3

Question 2

Your project team has been assigned to handle the Beamscope's project, duration is 24-months with a high-budget project. The user requirements for the system are still uncertain but the company has committed to a few departmental managers to be highly involved and provide consultation for this project. In addition, the scale of the project is large and it entails the use of a new technology and programming language that your team is unfamiliar with.

Consider which software process model would you propose for the above project: *Rapid Application Development* or *Spiral model*? Explain your answer.

Propose the Spiral Model for the Beamscope's project.

1. Uncertain Requirements:

Since the user requirements are still unclear, the Spiral Model is suitable because it supports iterative refinement through repeated cycles. Each iteration allows for gathering feedback and evolving the requirements, reducing the risk of misunderstanding user needs.

2. Large-Scale & High-Budget Project:

The Spiral Model is designed for complex and high-risk projects. It provides strong risk management mechanisms through each phase (planning, risk analysis, engineering, and evaluation), making it ideal for large and expensive projects where failure is costly.

3. New Technology & Language:

Because the team is unfamiliar with the technology and language, the risk is high. Spiral allows for early identification and mitigation of technical risks by incorporating prototyping and feasibility studies into each iteration.

4. Long Duration (24 months):

The extended timeline supports the Spiral Model's need for multiple iterative cycles. This ensures continuous improvement and risk assessment throughout the project.

5. Active Stakeholder Involvement:

Departmental managers are involved in consultations, which aligns with the evaluation phase of Spiral where continuous stakeholder feedback is integrated into the process.

Question 3

Given the following information in table 1 below:

Table 1: Status of Project A

(Refer to Appendix 2.2)

Activity	Estimated Duration (month)	Estimated Cost (RM)	Status	Earned Value (EV)
A	1	20,000	50%	
B	3	20,000	80%	
C	2	30,000	80%	
D	4	40,000	70%	
E	4	45,000	60%	
F	4	25,000	0%	
G	2	20,000	0%	
Total	20	200,000	-	

Assume that project A is now in its 10th month. As of today, you have paid out RM 103,000 to the contractors. Contractors are those who performed the tasks shown in Table 1. The budget at completion for project A is RM200,000. Show the workings in your calculation. Formula: $SV = EV - PV$, $CV = EV - AC$.

Activity	Estimated Duration (month)	Estimated Cost (RM)	Status	Earned Value (EV)
A	1	20,000	50%	
B	3	20,000	80%	
C	2	30,000	80%	
D	4	40,000	70%	
E	4	45,000	60%	
F	4	25,000	0%	
G	2	20,000	0%	
Total	20	it 200,000	-	

- a) Calculate the following values based on information provided in Table 1:
- | | |
|--------------------------|----------------------|
| - Planned Value (PV) | - Earned Value (EV) |
| - Schedule Variance (SV) | - Cost Variance (CV) |

Based on the calculated results, provide comments with an analysis on performance of this project.

- b) Based on the results calculated in part a), explain ONE (1) corrective action to ensure the project's success.

Team 1

a)

Activity	Estimated Duration (month)	Estimated Cost (RM)	Status	Earned Value (EV)
A	1	20,000	50%	$20000 \times 50\% = 10,000$
B	3	20,000	80%	$20000 \times 80\% = 16,000$
C	2	30,000	80%	$30000 \times 80\% = 24,000$
D	4	40,000	70%	$40000 \times 70\% = 28,000$
E	4	45,000	60%	$45000 \times 60\% = 27,000$
F	4	25,000	0%	$25,000 \times 0\% = 0$
G	2	20,000	0%	$20,000 \times 0\% = 0$
Total	20	200,000	-	$10000 + 16000 + 24000 + 28000 + 27000 + 0 + 0$ $= 105,000$

$PV = 20,000 + 20,000 + 30,000 + 40,000$
 $= \text{RM } 110,000 \text{ (10 months)}$

$EV = (20k * 50\%) + (20k * 80\%) + (30k * 80\%) + (40k * 70\%) + (45k * 60\%)$
 $= 10,000 + 16,000 + 24,000 + 28,000 + 27,000$
 $= \text{RM } 105,000$

$SV = EV - PV$ $= 105,000 - 110,000$ $= -\text{RM } 5,000$	$CV = EV - AC$ $= 105,000 - 103,000$ $= \text{RM } 2,000$
--	---

$\therefore$ SV is negative means that the project is behind schedule (In the 10th month, actual value of work completed is < value of work planned to be completed)

$\therefore$ CV is positive, it means that the project is under budget, the amount paid is less than actual value of work completed.

b)

adding more staff

Explanation:

When the Schedule Variance (SV) is negative, it means the project is behind schedule — the amount of work completed so far is less than what was planned.

By adding more staff, the project team gains additional workforce, allowing more tasks to be performed in parallel or speeding up delayed activities. This helps to accelerate progress and bring the project back on track.

Example:

If coding is behind schedule due to insufficient developers, assigning two more programmers can help finish modules faster and close the schedule gap.

Team 2

Question 3

Given the following information in table 1 below:

Table 1: Status of Project A

(Refer to Appendix 2.2)

Activity	Estimated Duration (month)	Estimated Cost (RM)	Status	Earned Value (EV)
A	1	20,000	50%	
B	3	20,000	80%	
C	2	30,000	80%	
D	4	40,000	70%	
E	4	45,000	60%	
F	4	25,000	0%	
G	2	20,000	0%	
Total	20	200,000	-	

Assume that project A is now in its 10th month. As of today, you have paid out RM 103,000 to the contractors. Contractors are those who performed the tasks shown in Table 1. The budget at completion for project A is RM200,000. Show the workings in your calculation. Formula: $SV = EV - PV$, $CV = EV - AC$.

25. Class Comments

- a) Calculate the following values based on information provided in Table 1:
- Planned Value (PV)
 - Earned Value (EV)
 - Schedule Variance (SV)
 - Cost Variance (CV)

Based on the calculated results, provide comments with an analysis on performance of this project.

- b) Based on the results calculated in part a), explain ONE (1) corrective action to ensure the project's success.

a)

Activity	Estimated Duration (month)	Estimated Cost (RM)	Status	Earned Value (EV)
A	1	20,000	50%	$20,000 * 50\% = 10,000$
B	3	20,000	80%	$20,000 * 80\% = 16,000$
C	2	30,000	80%	$30,000 * 80\% = 24,000$
D	4	40,000	70%	$40,000 * 70\% = 28,000$
E	4	45,000	60%	$45,000 * 60\% = 27,000$

F	4	25,000	0%	25,000 * 0% = 0
G	2	20,000	0%	20,000 * 0% = 0
Total	20	200,000	-	10,000 + 16,000 + 24,000 + 28,000 + 27,000 + 0 + 0 = RM 105,000

PV

Activity A+B+C+D = 10 months

= 20,000 + 20,000 + 30,000 + 40,000

= RM 110,000

AC = RM 103,000

EV = 10,000 + 16,000 + 24,000 + 28,000 + 27,000

= RM 105,000

SV = EV - PV

= 105,000 - 110,000

= RM -5,000 (negative **behind schedule**)

CV = EV - AC

= 105,000 - 103,000

= RM 2,000 (**under budget**)

- SV is negative (-RM 5000) means that the project is behind schedule (In the 10th month, the actual value of work completed is less than the value of work planned to be completed)
- CV is positive (RM 2000), which means that the project is under budget. In the 10th month, the actual value of work completed is more than the amount paid

b) Adding More Staff

- Adding more personnel can help complete the tasks faster, especially if the project is behind schedule due to a lack of resources. By increasing the workforce, the tasks can be handled simultaneously by more people, accelerating progress.

Team 3

Table 1: Status of Project A

(Refer to Appendix 2.2)

Activity	Estimated Duration (month)	Estimated Cost (RM)	Status	Earned Value (EV)
A	1	20,000	50%	$20,000 \times 50\% = 10,000$
B	3	20,000	80%	$20,000 \times 80\% = 16,000$
C	2	30,000	80%	$30,000 \times 80\% = 24,000$
D	4	40,000	70%	$40,000 \times 70\% = 28,000$
E	4	45,000	60%	$45,000 \times 60\% = 27,000$
F	4	25,000	0%	$25,000 \times 0\% = 0$
G	2	20,000	0%	$20,000 \times 0\% = 0$
Total	20	200,000	-	105,000

$$a. PV = 20,000 + 20,000 + 30,000 + 40,000 \\ = RM 110,000$$

$$SV = EV - PV \\ = 105,000 - 110,000 \\ = -RM 5,000$$

$\therefore SV$ is - RM 5,000, it means that

the project is significantly behind schedule. May cause poor time management, underestimated task durations, or lack of manpower.

$$EV = 10,000 + 16,000 + 24,000 + 28,000 + 27,000 \\ = RM 105,000$$

$$CV = EV - AC \\ = 105,000 - 103,000 \\ = RM 2,000$$

$\therefore CV$ is RM 2,000, showing the project is significantly under budget.

b. Corrective Action: Improving Methods of Working (behind schedule and cost is under control)

- applying better task tracking tools or project management software
- conduct daily stand-up meetings to identify blockers
- Revisiting dependencies and eliminating inefficiencies.

Team 4

Activity	Estimated Duration (month)	Estimated Cost (RM)	Status	Earned Value (EV)
A	1	20,000	50%	$20,000 * 0.5 = 10,000$
B	3	20,000	80%	$20,000 * 0.8 = 16,000$
C	2	30,000	80%	$30,000 * 0.8 = 24,000$
D	4	40,000	70%	$40,000 * 0.7 = 28,000$
E	4	45,000	60%	$45,000 * 0.6 = 27,000$
F	4	25,000	0%	$25,000 * 0.0 = 0$
G	2	20,000	0%	$20,000 * 0.0 = 0$
Total	20	200,000	-	105,000

$$PV = 40,000 + 30,000 + 20,000 + 20,000 = 110,000$$

$$EV = 10,000 + 16,000 + 24,000 + 28,000 + 27,000 + 0 + 0 = 105,000$$

$$SV = EV - PV = 105,000 - 110,000 = -5,000$$

$$CV = EV - AV = 105,000 - 55,000 = -50,000$$

Question 4

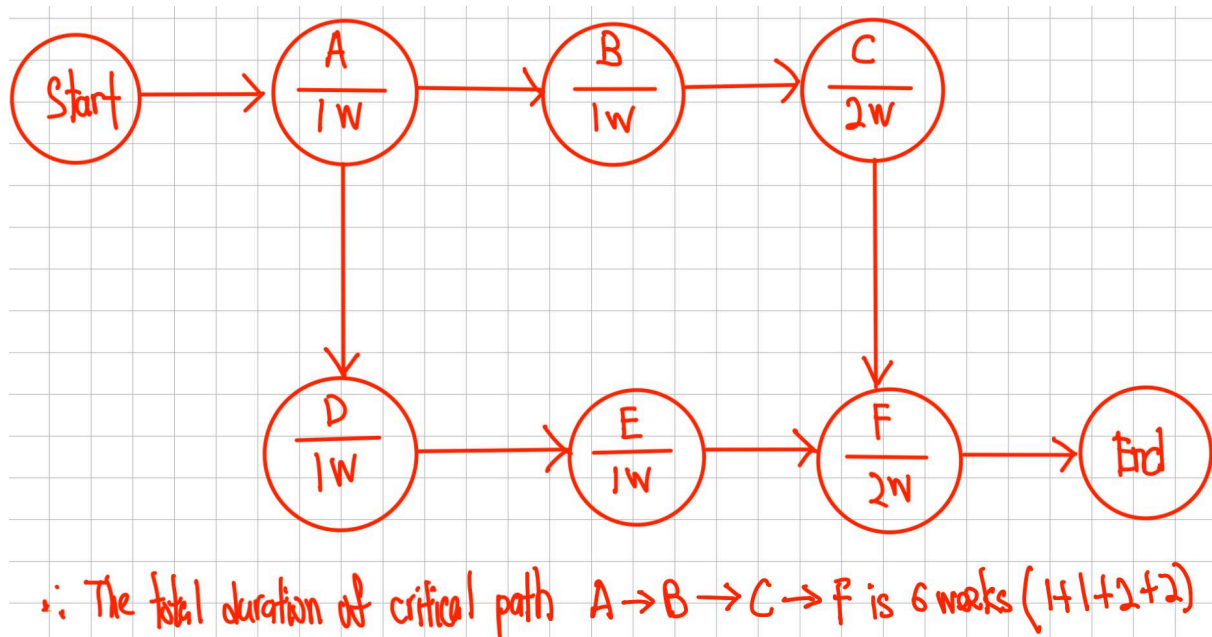
Question 4

A network diagram can be used to illustrate the interdependencies between tasks in a project. Table 1 shows Project tasks, duration and its predecessor.

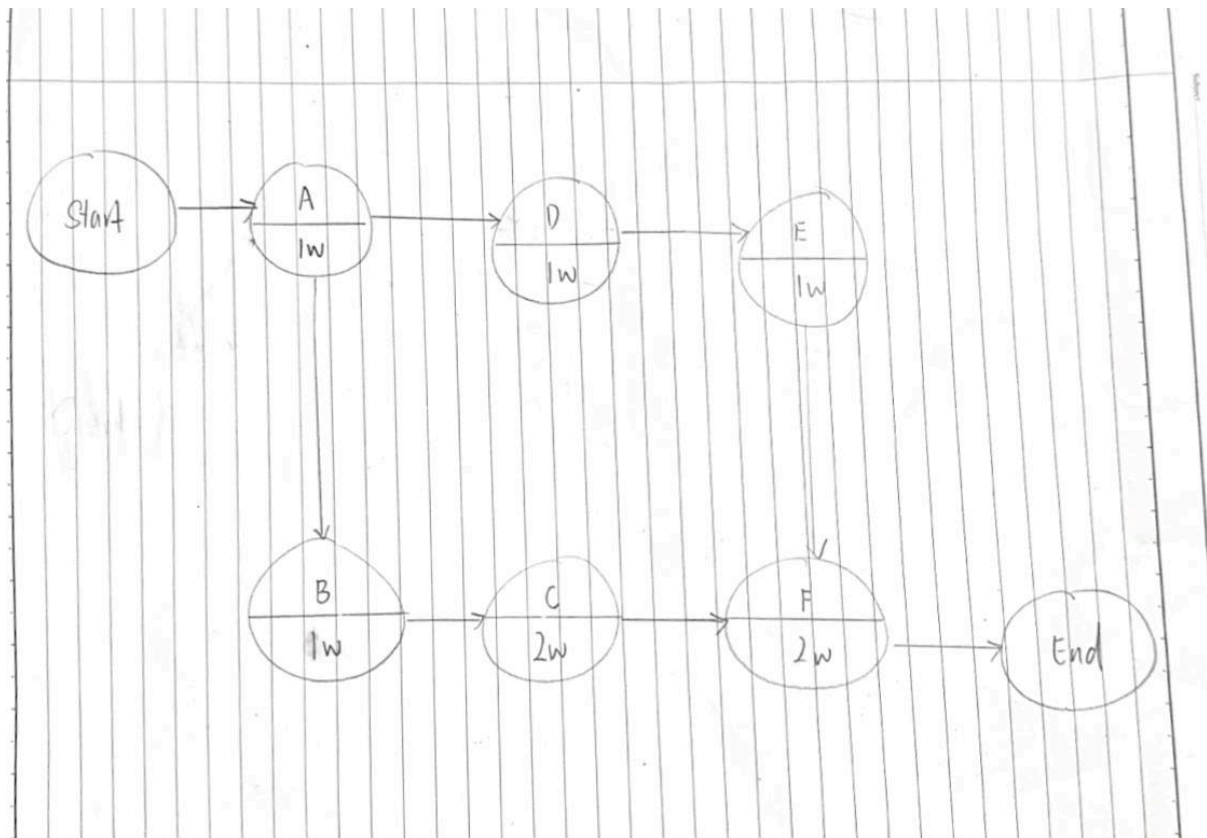
Task ID	Task Name	Duration (in week)	Predecessor
A	Collect requirements	1	None
B	Design online module	1	A
C	Code online module	2	B
D	Design offline module	1	A
E	Code offline module	1	D
F	Testing online and offline module	2	C, E

- Create a network diagram based on information provided in Table 1.
- Based on the answer given in part a), identify the critical path based on the task ID and specify the total duration needed to complete the project.

Team 1



Team 2



$A \rightarrow B \rightarrow C \rightarrow F$

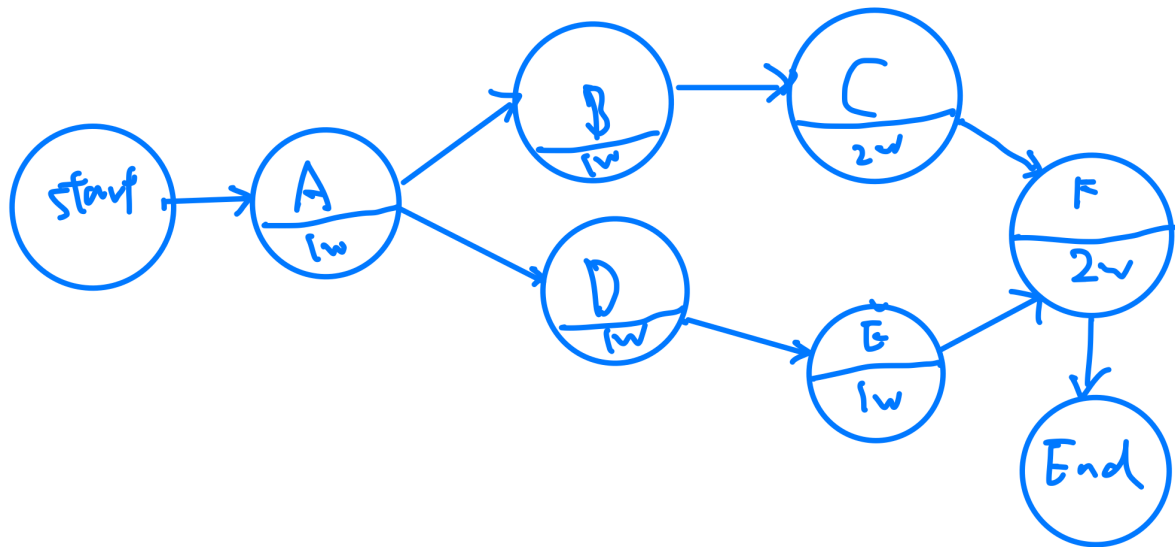
$= 1 + 1 + 2 + 2 = 6$ weeks

The critical path is the one with the longest duration, which is:

Critical Path: $A \rightarrow B \rightarrow C \rightarrow F$

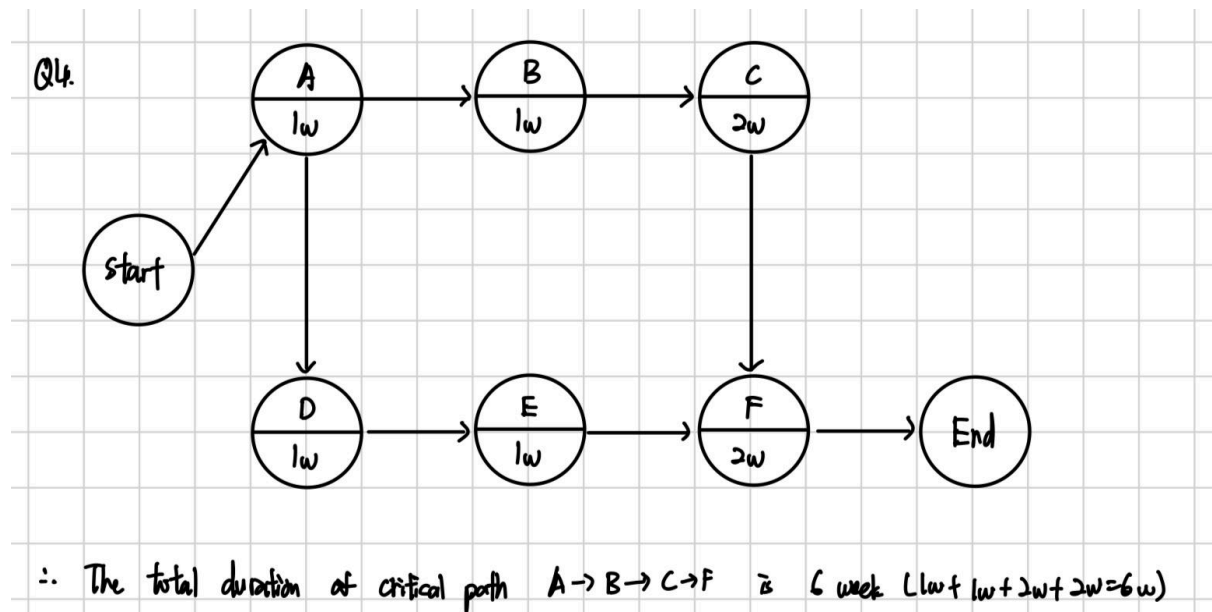
Total Project Duration: 6 weeks

Team3



The total duration of critical path A → B → C → F is 6 week ($1w+1w+2w+2w=6w$)

Team 3 (Tan)



Team 4

Question 5 (Team 1)

Gantt charts are used to illustrate the sequence of activities in a project. Figure 1 is an example of a Gantt chart. Redraw the Gantt chart in Figure 1 using the predecessor information provided in Table 1 to illustrate the interdependencies between project activities. Assuming the project starts in June 2024.

Figure 1: Example of a Gantt chart

Task ID	Task Name	Jun '24	Jul '24	Aug '24	Sep '24	Oct '24
1	Analyse Requirements					
2	Design online module					
3	Code online module					
4	Design offline module					
5	Code offline module					
6	Test online and offline module					

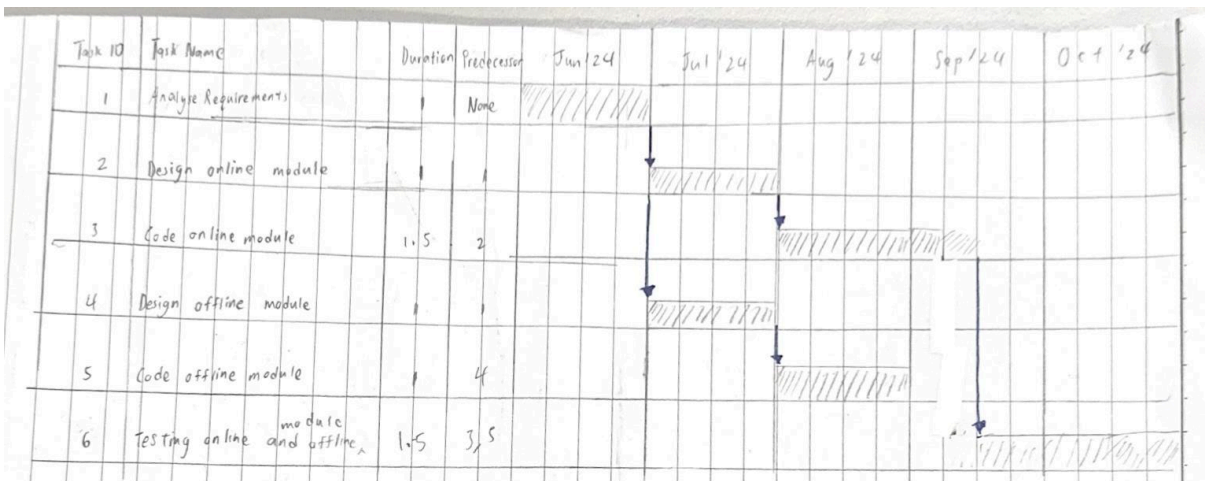
Table 1: Project activity and its predecessor

Task ID	Task Name	Duration (in month)	Predecessor
1	Analyze requirements	1	None
2	Design online module	1	1
3	Code online module	1.5	2
4	Design offline module	1	1
5	Code offline module	1	4
6	Testing online and offline module	1.5	3, 5

Identify the month for the task ID 6 will be completed.

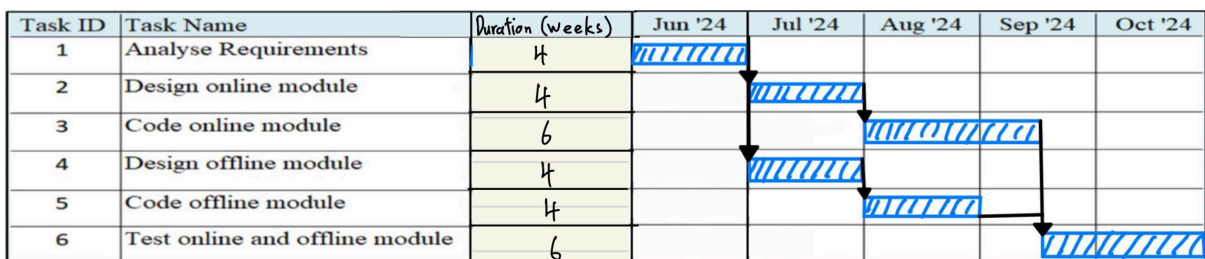
Team 2

Answer:



The month for task ID 6 will be completed at the end of October.

Answer:



The month for task ID 6 will be completed at End of October.

Q5 Team 3

Task ID	Task Name	Duration (months)	Predecessor	Jun 2024	July 24	Aug 24	Sept 24	Oct 24
1	Analyze requirements	1	None					
2	Design online module	1	1					
3	Code Online module	1.5	2					
4	Design Offline module	1	1					
5	Code Offline module	1	4					
6	Testing online and offline module	1.5	3,5					

∴ The month for task ID 6 will be completed is October.

Q5 Team 5

Task ID	Task Name	Duration (in month)	Predecessor	Jun 24	Jul 24	Aug 24	Sep 24	Oct 24
1	Analyze Requirements	1	None					
2	Design online module	1	1					
3	Code online module	1.5	2					
4	Design offline module	1	1					
5	Code offline module	1	4					
6	Test online & offline module	1.5	3,5					

∴ The task ID 6 will be completed in October 24.